



SAVEETHA
SCHOOL OF ENGINEERING
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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

REVIEW 2-PRESENTATION

Predictive Wildfire Spread Simulator with Evacuation Route Optimizer

CSA0915– Java Programming

Presented By:

S.K.Deepa Arathi (192571020)

Lochana Sri R S (192521348)

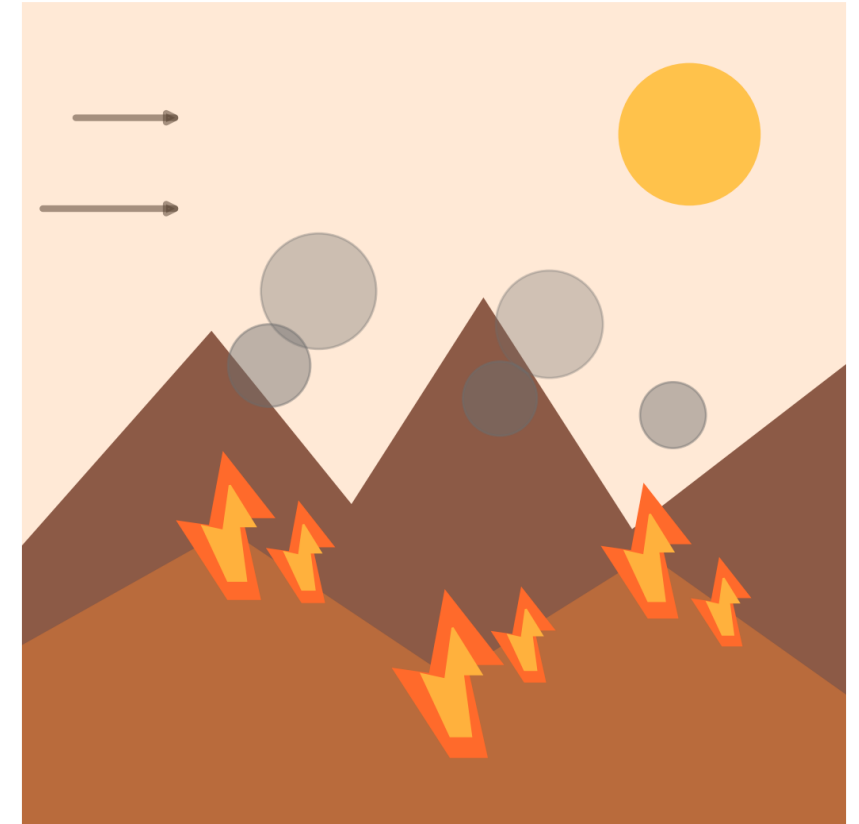
Nitya Priya P M (192572086)

Guided By:

Dr.E.Meganathan

INTRODUCTION

- Wildfires are among the most destructive natural disasters, causing loss of life, forests, and property.
- Climate change, high temperatures, strong winds, and dry vegetation have increased wildfire frequency and intensity.
- Early prediction of wildfire spread is essential to reduce damage and improve emergency response.
- FireWatch is an intelligent system that predicts the direction and speed of wildfire spread using environmental data.
- The system considers factors such as wind speed, temperature, humidity, vegetation, and terrain.



PROBLEM STATEMENT

- Wildfires spread rapidly due to changing weather conditions, making them difficult to predict.
- Existing wildfire monitoring systems mainly detect fires after they have already started, limiting response time.
- Delayed prediction and evacuation can lead to significant loss of lives, property, forests, and wildlife.
- Emergency responders often face challenges in identifying the safest and fastest evacuation routes.
- Real-time environmental factors such as wind speed, temperature, humidity, and terrain are not always integrated into decision-making.



PROPOSED SYSTEM

- To develop an intelligent system for predicting wildfire spread.
- To analyze wind, temperature, humidity, vegetation, and terrain for better accuracy.
- To simulate wildfire spread direction and speed in real time.
- To identify the safest and shortest evacuation routes.
- To provide early warnings for emergency response teams.
- To reduce loss of life, property, and forests through timely evacuation.
- To improve disaster management via combined prediction and route planning.



MODULES

Module 1: Data Collection & Wildfire Prediction

Module 2: Wildfire Spread Simulation

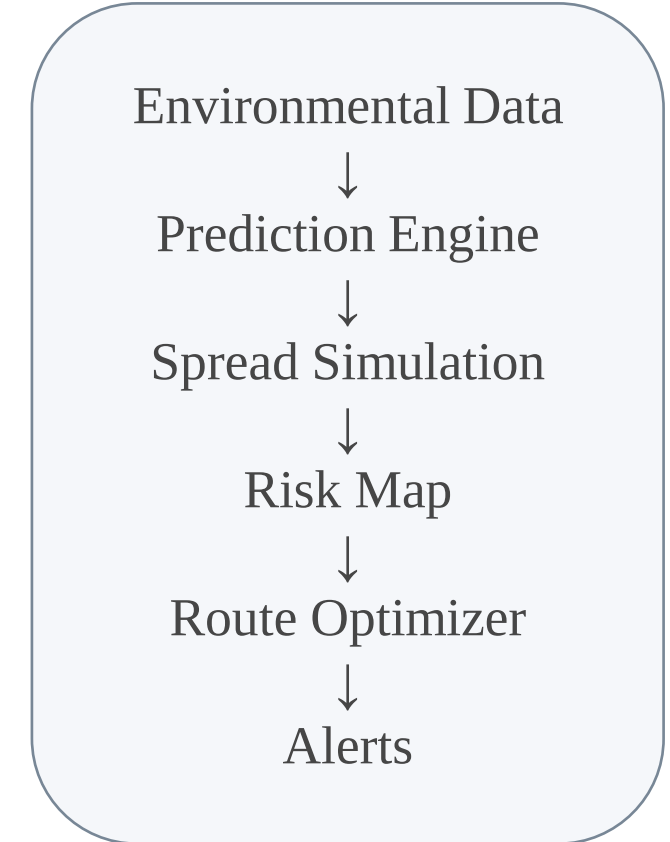
Module 3: Evacuation Route Optimization & Alert System



SYSTEM ARCHITECTURE



- FireWatch combines environmental data analysis, wildfire prediction, spread simulation, evacuation route optimization, and alert generation.
- Input data includes weather conditions, vegetation, terrain, road information, safe locations, and the detected fire location.
- Each stage produces information that becomes the input for the next stage.



MODULE 1: DATA COLLECTION & WILDFIRE PREDICTION

- Collect temperature, humidity, wind speed and direction, vegetation condition, terrain slope, and current fire-location data.
- Preprocess the data by checking missing values and converting all inputs into a suitable format.
- Analyze the environmental conditions to estimate the likely direction and relative speed of wildfire spread.
- Output: predicted spread parameters and an initial risk estimate.

MODULE 1 – ENVIRONMENTAL DATA & WILDFIRE RISK PREDICTION

ENTER ENVIRONMENTAL DATA

LOCATION NAME
e.g. Pine Ridge

TEMPERATURE (°C)
32

HUMIDITY (%)
35

WIND SPEED (KM/H)
20

WIND DIRECTION
N

RAINFALL (MM, LAST 24H)
0

VEGETATION TYPE

RISK RESULT

64.03

HIGH

Pine Ridge

CONTRIBUTING FACTORS

Temperature (25%)	17.78
Wind (25%)	6.25
Humidity (20%)	13.0
Vegetation (15%)	13.5
Terrain (15%)	13.5
Rainfall damping	-0.0
Existing fire intensity	0.0

MODULE 1 – WORKING



- Step 1: Receive environmental and geographic inputs.
- Step 2: Validate and preprocess the collected data.
- Step 3: Analyze factors influencing fire behaviour.
- Step 4: Predict the likely spread direction and speed.
- Step 5: Pass the prediction to the simulation module.

Input

Process

Analysis

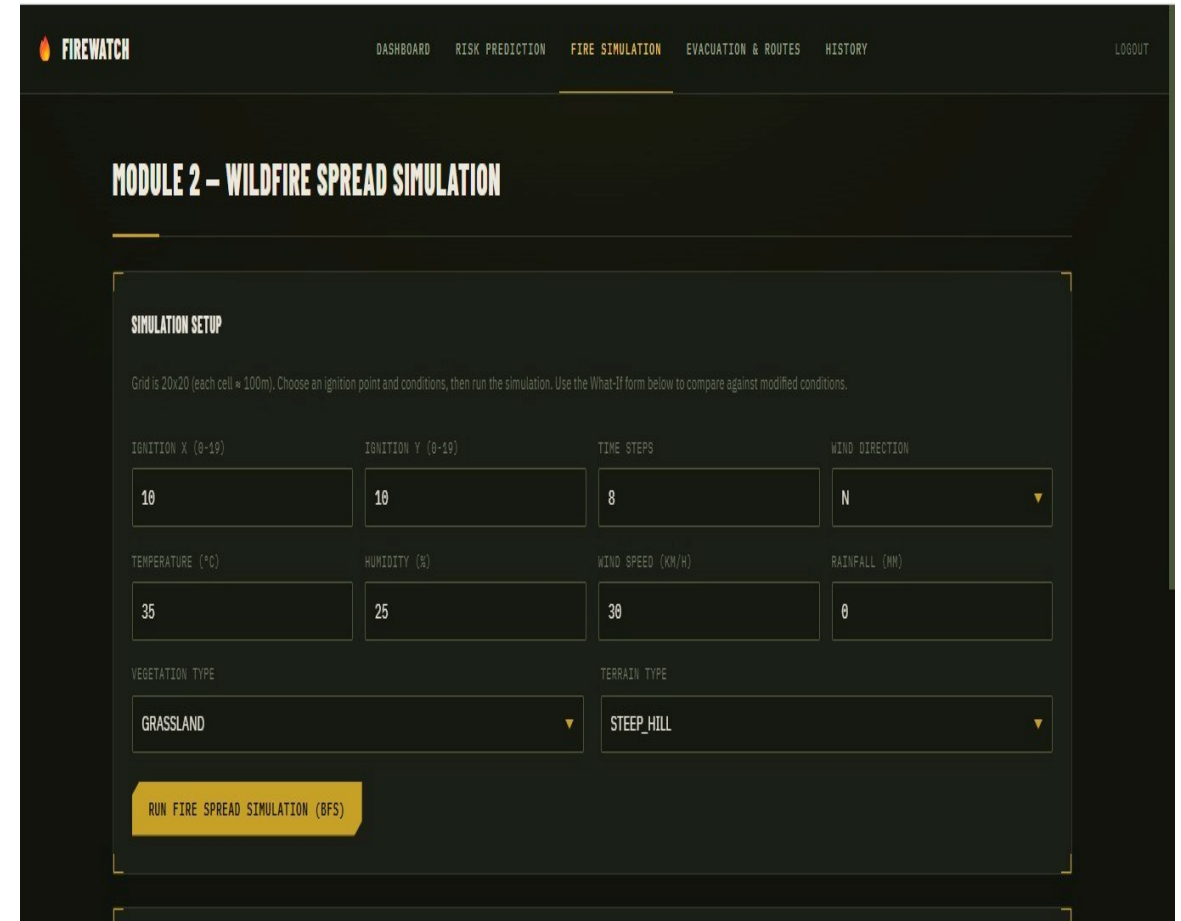
Output

Update

MODULE 2: WILDFIRE SPREAD SIMULATION



- Represent the affected area using map cells or geographic zones.
- Start the simulation from the detected wildfire location.
- Update neighbouring zones according to wind, humidity, vegetation, terrain, and predicted spread behaviour.
- Generate a time-based view showing how the risk area may expand.
- Output: a dynamic wildfire risk map.



The screenshot shows the 'FIREWATCH' web application interface. The top navigation bar includes 'DASHBOARD', 'RISK PREDICTION', 'FIRE SIMULATION' (highlighted), 'EVACUATION & ROUTES', 'HISTORY', and 'LOGOUT'. The main heading is 'MODULE 2 – WILDFIRE SPREAD SIMULATION'. Below this is a 'SIMULATION SETUP' section with a note: 'Grid is 20x20 (each cell = 100m). Choose an ignition point and conditions, then run the simulation. Use the What-If form below to compare against modified conditions.'

IGNITION X (0-19)	IGNITION Y (0-19)	TIME STEPS	WIND DIRECTION
10	10	8	N

TEMPERATURE (°C)	HUMIDITY (%)	WIND SPEED (KM/H)	RAINFALL (MM)
35	25	30	0

VEGETATION TYPE	TERRAIN TYPE
GRASSLAND	STEEP_HILL

RUN FIRE SPREAD SIMULATION (BFS)

MODULE 2 – WORKING



- Step 1: Initialize the map and fire source location.
- Step 2: Divide the region into zones or cells.
- Step 3: Calculate the influence of environmental conditions.
- Step 4: Update the predicted fire boundary for each time interval.
- Step 5: Mark zones according to their risk level.

Input

Process

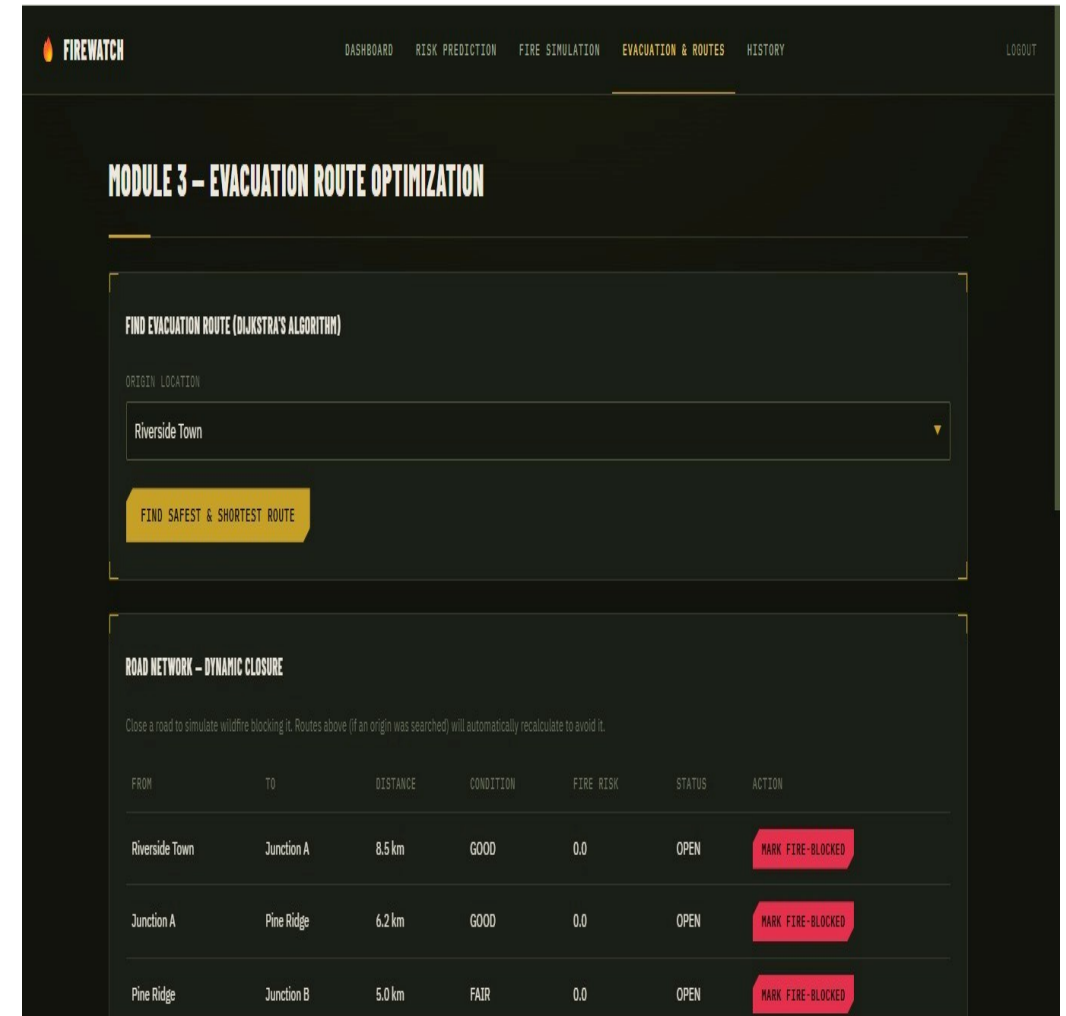
Analysis

Output

Update

MODULE 3: EVACUATION ROUTE OPTIMIZATION & ALERT SYSTEM

- Combine the wildfire risk map with roads, exits, and safe shelter locations.
- Avoid or assign high cost to routes passing through predicted high-risk zones.
- Use graph-based algorithms such as Dijkstra's or A* to identify safer routes.
- Recalculate the route when the predicted wildfire situation changes.
- Generate alerts containing risk status and recommended evacuation information.



FIREWATCH DASHBOARD RISK PREDICTION FIRE SIMULATION **EVACUATION & ROUTES** HISTORY LOGOUT

MODULE 3 – EVACUATION ROUTE OPTIMIZATION

FIND EVACUATION ROUTE (DIJKSTRA'S ALGORITHM)

ORIGIN LOCATION

Riverside Town

FIND SAFEST & SHORTEST ROUTE

ROAD NETWORK – DYNAMIC CLOSURE

Close a road to simulate wildfire blocking it. Routes above (if an origin was searched) will automatically recalculate to avoid it.

FROM	TO	DISTANCE	CONDITION	FIRE RISK	STATUS	ACTION
Riverside Town	Junction A	8.5 km	GOOD	0.0	OPEN	MARK FIRE-BLOCKED
Junction A	Pine Ridge	6.2 km	GOOD	0.0	OPEN	MARK FIRE-BLOCKED
Pine Ridge	Junction B	5.0 km	FAIR	0.0	OPEN	MARK FIRE-BLOCKED

MODULE 3 – WORKING



- Step 1: Receive the latest risk map from the simulation module.
- Step 2: Identify the user's or affected area's starting point.
- Step 3: Evaluate available roads and remove unsafe paths.
- Step 4: Calculate a safer route to a safe destination.
- Step 5: Display the route and generate an alert.

Input

Process

Analysis

Output

Update

METHODOLOGY

- Data collection provides the environmental and geographic information.
- The prediction engine estimates the possible fire movement.
- The simulation module converts the prediction into a changing risk map.
- The route optimizer searches for a safer path using the current risk information.
- The alert system communicates the recommendation and updates it when new conditions are available.

Collect

Predict

Simulate

Risk Map

Optimize

Alert



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JAVA PROGRAMMING IMPLEMENTATION



- Java can implement the project using object-oriented programming and modular classes.
- Suggested classes: WeatherData, FireCell, FirePredictor, SpreadSimulator, RiskMap, Graph, RouteOptimizer, and AlertService.
- Arrays or ArrayLists can represent simulation data and geographic zones.
- Graph data structures can represent roads and locations.
- Dijkstra's or A* algorithm can be used for evacuation route calculation.



REAL-LIFE EXAMPLE SCENARIO



- Assume a wildfire is detected near a forest boundary close to a residential area.
- Strong wind and low humidity indicate that the fire may spread more rapidly in a particular direction.
- The simulator marks the predicted path as a higher-risk zone.
- The route optimizer avoids roads entering the high-risk area and searches for an alternative route.
- Emergency teams can use the result to support evacuation decisions.



EXPECTED OUTPUTS



- Predicted wildfire spread direction and relative speed.
- Dynamic wildfire risk map.
- Identification of areas that may become unsafe.
- Recommended evacuation route based on current predicted risk.
- Alert messages for emergency response and evacuation.
- Updated recommendations when input conditions change.

 Risk Map

 Route

 Alert

ADVANTAGES

- Combines prediction, simulation, route planning, and alerts in one system.
- Provides earlier decision-support information for emergency situations.
- Supports route selection using predicted wildfire risk.
- Modular design makes testing and future expansion easier.
- Demonstrates a real-world application of Java programming and graph algorithms.



LIMITATIONS



Engineer to Excel Prediction quality depends on the accuracy and availability of input data.

- Actual wildfire behaviour can change rapidly because environmental conditions are dynamic.
- Real-world deployment requires reliable maps, weather feeds, and continuous updates.
- Recommended routes must be validated with real road and emergency information.
- The current academic prototype should be tested extensively before operational use.



Data Quality
Real-time Changes
Validation Needed



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FUTURE ENHANCEMENTS



- Integrate real-time weather data and sensor or satellite observations.
- Use machine learning models trained on historical wildfire information.
- Add GIS maps with live road closures and safe shelter details.
- Develop a mobile application for location-aware guidance.
- Include traffic conditions and multiple evacuation-route options.
- Create an emergency authority dashboard for monitoring the situation.



AI + GIS
Real-time Data
Mobile App

CONCLUSION

- FireWatch integrates wildfire prediction, spread simulation, and evacuation route optimization into a single workflow.
- The three modules provide a clear structure for developing the system.
- Environmental information is converted into a predicted risk map that supports route planning.
- The project provides a practical Java-based application involving object-oriented programming, data structures, and graph algorithms.

 Dashboard

 Live Map

 Fire Prediction

 Simulation

 Evacuation Routes

 Alerts

 Reports

 Settings

 Logout

EMERGENCY

 112

Fire Control Room

 **ACTIVE FIRES**
3
High Risk

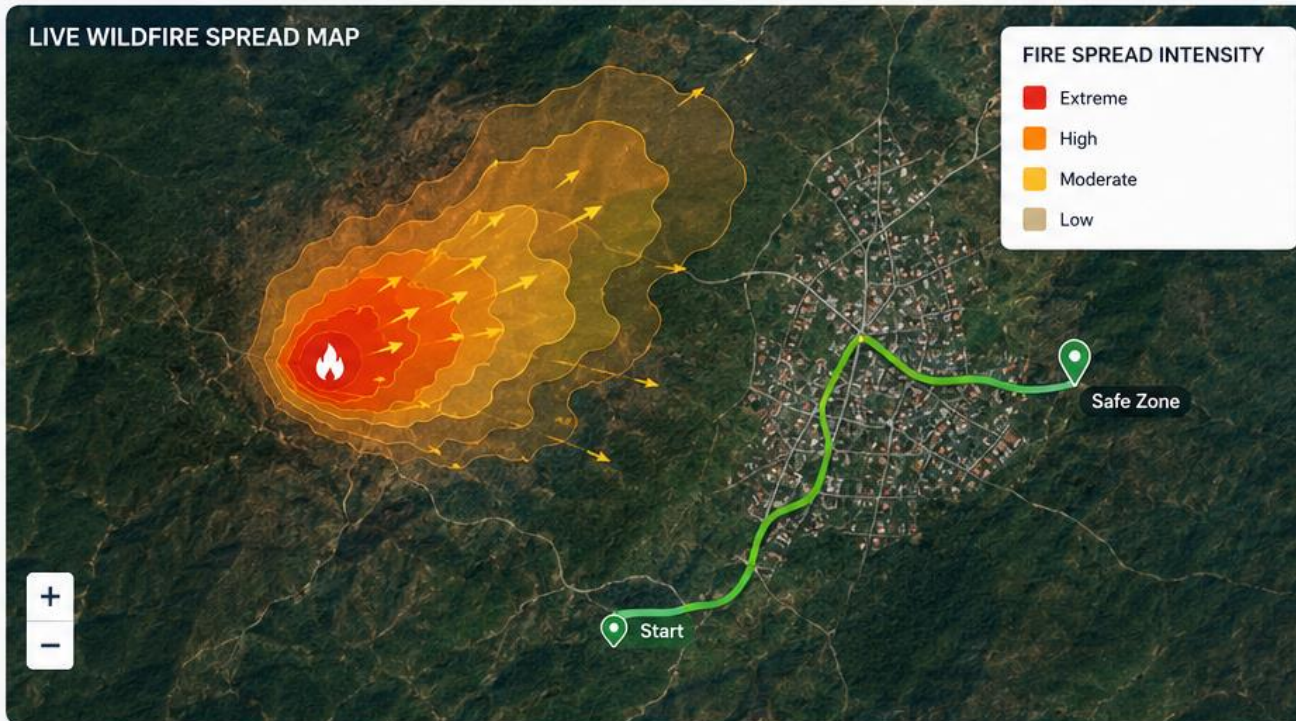
 **TEMPERATURE**
32°C
High

 **WIND SPEED**
25 km/h
NW

 **HUMIDITY**
28%
Low

 **VEGETATION**
Dry
High Risk

LIVE WILDFIRE SPREAD MAP



EVACUATION ROUTE

Recommended Route

Distance **8.7 km**
Estimated Time **18 min**
Safety Level **High**

[View Full Route](#)

ALERTS & NOTIFICATIONS

-  High fire intensity detected near North Zone 10:32 AM
-  Strong winds towards NE expected in 2 hours 10:15 AM
-  Evacuation advised for residents in West Zone 09:48 AM

[View All Alerts](#)

FIRE RISK INDEX



WEATHER CONDITIONS

 **32°C**
Temperature

 **25 km/h**
Wind Speed

 **28%**
Humidity

 **0 mm**
Rainfall

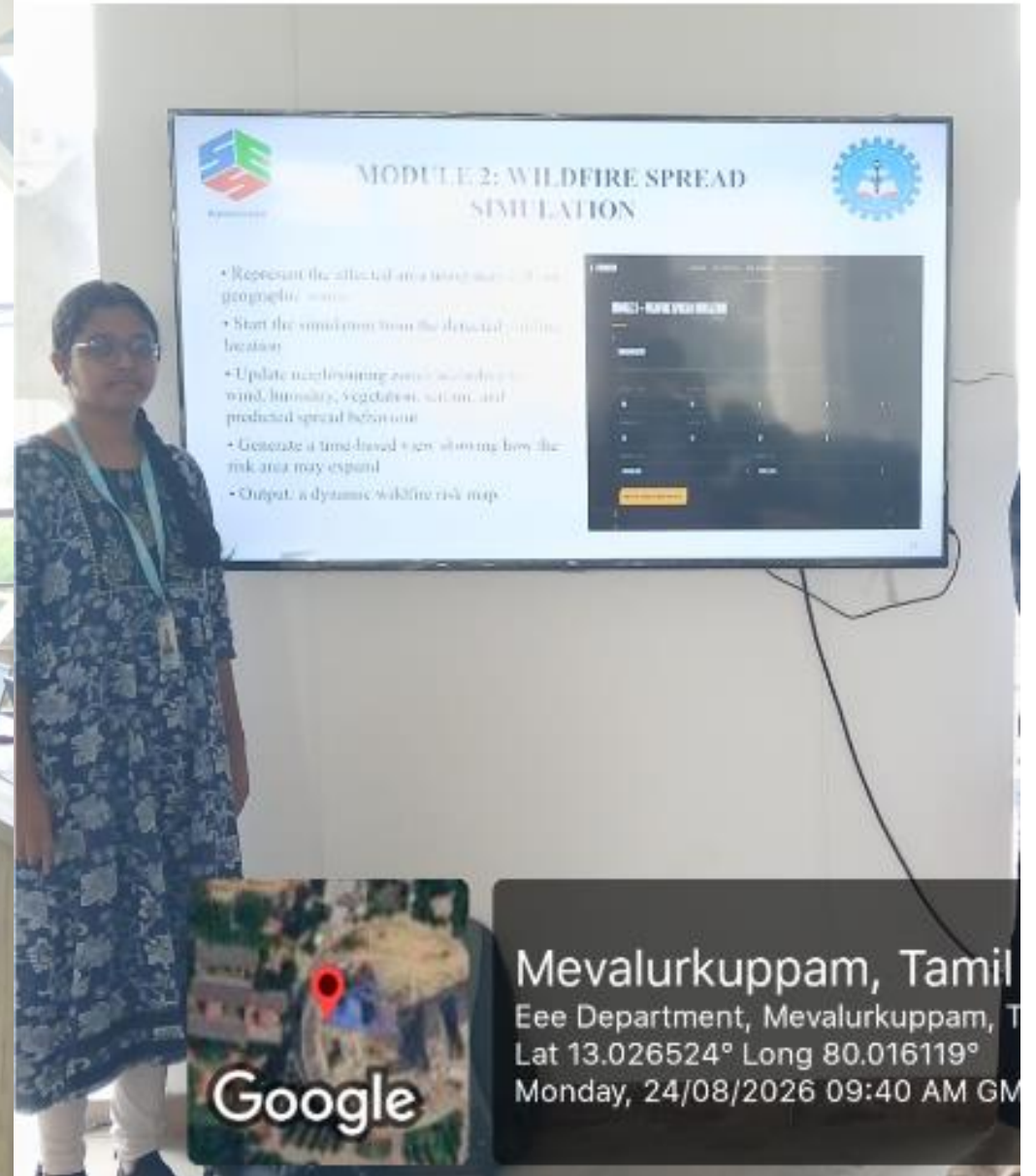
PREDICTION SUMMARY

Fire is predicted to spread towards NE direction with high intensity in next 6 hours.

Affected Area (Est.)
1250 ha



GEO TAG PHOTO



THANK
YOU